

OCR A-Level Chemistry A Paper 2: Student Strategy Guide

Based on 9 Past Papers + 8 Examiner Reports (2017-2025)

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Important disclaimer This guide is based on HSJ Tuition's independent analysis of all publicly available OCR Chemistry A Paper 2 (H432/02) question papers and examiner reports from 2017 to 2025. We are highly confident in the patterns and advice presented here: they are grounded in nine years of examiner evidence, not guesswork. However, **exam content cannot be predicted or guaranteed**. OCR can set any topic from the specification at any time. This guide is a revision strategy tool designed to help you prioritise your time and eliminate common errors. Use it alongside full specification coverage, not instead of it. HSJ Tuition accepts no responsibility for topics not appearing or appearing differently than anticipated.

WHY THIS GUIDE EXISTS

This guide was built by analysing every OCR Chemistry A Paper 2 (H432/02) from 2017 through 2025, plus every examiner report published for this paper. It tells you **exactly where students lose marks**, not based on opinion, but on what examiners have flagged year after year.

WHAT PAPER 2 ASSESSES

H432/02: Synthesis and Analytical Techniques. Written exam, 2 hours 15 minutes, 100 marks (37% of the A-level).

Section	Format	Marks	Time
A	15 multiple choice questions	15	~20 mins
B	Short and extended answer questions	85	~115 mins

Section B always includes **2 Level of Response questions** (marked with *, worth 6 marks each, using Level 3/2/1 banding).

Paper 2 tests: **organic chemistry**, **analytical techniques** (IR, NMR, mass spec), and **relevant practical skills**. Physical chemistry topics (rates, equilibria, enthalpy, electrochemistry, acids/bases, transition metals) are assessed on Paper 1 and Paper 3 only. Some basic moles/stoichiometry and intermolecular forces appear when set in an organic context.

PART 1: WHAT TO REVISE FIRST

These topics appear **every single year** (9/9 papers). They are all top priority:

Priority	Topic	Frequency	Marks per paper
1	Carboxylic acids and esters (formation, hydrolysis, naming)	9/9	5-10
2	Alcohols (oxidation, elimination, classification)	9/9	5-10
3	Combined analytical / structure determination	9/9	6-10
4	Multi-step organic synthesis (with % yield)	9/9	5-8
5	Aldehydes/ketones (tests, NaBH ₄ reduction, mechanism)	9/9	4-8
6	IR spectroscopy	9/9	2-4
7	Mass spectrometry	9/9	2-4
8	Moles / stoichiometry calculations	9/9	2-5

These also have **high frequency** (8/9 or 7/9 papers):

Priority	Topic	Frequency
9	Chemical tests (2,4-DNP, Tollens', bromine water)	8/9
10	Electrophilic addition (alkene + HBr/Br ₂)	8/9
11	Nucleophilic addition to carbonyls (HCN/NaBH ₄)	8/9
12	Nucleophilic substitution (haloalkane)	8/9
13	¹ H NMR	8/9
14	¹³ C NMR	8/9
15	Nitration mechanism	7/9

Priority	Topic	Frequency
16	Amino acids (optical isomerism, reactions, peptides)	7/9
17	Free radical substitution	7/9
18	Percentage yield	7/9
19	Phenol reactivity	7/9
20	Condensation polymers (polyesters)	7/9
21	Acyl chlorides	7/9

Do not skip priorities 1-8. These are guaranteed content. Together they account for 35-55 marks per paper.

PART 2: THE GOLDEN RULES

These errors are flagged in **multiple examiner reports**. They cost marks every year.

Rule 1: Curly arrows start from lone pairs or bonds, never from atoms

"The curly arrow from the hydroxide ion does not involve either the lone pair or minus sign on the O atom" (2018 ER). An arrow from an atom scores zero for that step. Every year.

Rule 2: Draw structures with correct hydrogen count

"Candidates need to make sure that when they draw structures of molecules, these must have the correct connectivity of atoms" (2020 ER). The most frequently cited error across all years. Count your hydrogens.

Rule 3: Do not mix skeletal and displayed formulae

"Candidates who drew structures that combined skeletal and displayed formulae produced ambiguous structures with missing hydrogen atoms" (2023 ER). Pick one format and stick to it.

Rule 4: Markownikoff is about carbocation stability, not product stability

"Some candidates explained the formation of the major organic product in terms of the stability of the product, rather than referring to the stability of the intermediate" (2021 ER). The more substituted carbocation is more stable. That determines the major product.

Rule 5: Optical isomers are "non-superimposable mirror images"

"It was still common to see incomplete responses such as 'mirror images' without reference to 'non-superimposable'" (2023 ER). Both words required.

Rule 6: Stereoisomers have the same structural formula, not molecular formula

"A significant number of candidates referred to stereoisomers as having the same molecular formula, rather than being more precise and stating that they are compounds with the same structural formula" (2019 ER).

Rule 7: CIP priority uses atomic number, not molecular mass

"A common misconception was to refer to the molecular mass of the groups instead of priority" (2018 ER). "A significant number of candidates described priority in terms of highest Mr or mass" (2024 ER). Atomic number only.

Rule 8: Curly arrows represent "a pair of electrons", not "electrons"

"Many candidates gave responses that lacked precision, such as 'movement of electrons'. Such responses did not receive credit as the number of electrons was not specified" (2019 ER). Must say "a pair of electrons."

Rule 9: Global warming and ozone depletion are different things

"Many candidates seemed to treat global warming and ozone depletion as the same thing" (2017 ER). "Many candidates believed there was a link between ultraviolet radiation and global warming" (2024 ER). Global warming = IR absorption by greenhouse gases. Ozone depletion = UV breaking C-Cl bonds in CFCs.

Rule 10: SOCl₂ converts carboxylic acids to acyl chlorides

"Common incorrect reagents included HCl and AlCl₃" (2018 ER). The reagent is SOCl₂ (thionyl chloride). Products: RCOCl + SO₂ + HCl.

PART 3: TOPIC-BY-TOPIC KEY MISTAKES

Electrophilic Addition

- Omitting partial charges (delta+ and delta-) on HBr/HCl in the first step of the mechanism.
- Omitting the charge on the carbocation intermediate.

- For Markownikoff: explain via the more stable (more substituted) carbocation, NOT by referring to product stability.
- For Br₂ addition: must show the induced dipole on Br₂ caused by the electron-rich pi bond.

Nucleophilic Substitution

- Arrow must start from the lone pair (or negative charge) on OH⁻/CN⁻, not from the atom itself (flagged 2018).
- Must show the leaving group (Cl⁻, Br⁻) as a product.
- NaCN in ethanol (not aqueous conditions) for haloalkane + cyanide: "Many included an acid catalyst or aqueous conditions (aq) so lost the mark" (2024 ER).

Free Radical Substitution

- Initiation must show homolytic fission (fish-hook arrows or stating "homolytic").
- Propagation must be balanced (atoms and radicals).
- Must know the limitations: mixture of products (mono/di/poly substitution), poor selectivity with branched chains.

Nucleophilic Addition to Carbonyls (HCN/NaBH₄)

- Reagent for HCN addition: NaCN + dilute H₂SO₄ (or KCN + acid). NOT just "CN⁻" alone.
- The mechanism requires: lone pair on CN⁻ (or H⁻) attacks delta+ carbonyl carbon; pi electrons pushed onto O forming O⁻; then protonation by HCN or H⁺.
- The carbonyl carbon is planar (sp²), so nucleophile attacks from either face with equal probability, giving a racemic mixture. Must state BOTH planar and equal attack.

Nitration Mechanism

- Electrophile formation: HNO₃ + H₂SO₄ -> NO₂⁺ + HSO₄⁻ + H₂O
- "Many candidates did not appreciate that two electrons are removed from the ring in the first stage" (2023 ER). Show the horseshoe arrow from the ring to NO₂⁺.
- The intermediate (arenium ion) must show: partial delocalisation over five carbons, + charge, H still attached to the carbon bearing NO₂⁺.
- H⁺ lost to restore aromaticity; taken by HSO₄⁻ to regenerate H₂SO₄ catalyst.
- Friedel-Crafts: must show AlCl₄⁻ with its charge, and the electrophile (RCO⁺ or R⁺) with its charge.

Carbonyl Tests

Reagent	Aldehyde result	Ketone result
2,4-DNP (Brady's)	Orange/yellow precipitate	Orange/yellow precipitate
Tollens' (ammoniacal AgNO ₃)	Silver mirror	No change
Fehling's (blue solution)	Brick-red precipitate	No change
Na ₂ CO ₃ / NaHCO ₃	No reaction (unless COOH present)	No reaction
K ₂ Cr ₂ O ₇ / H ₂ SO ₄	Orange to green	No change

2,4-DNP identifies C=O but does NOT distinguish aldehyde from ketone.

Acyl Chlorides

- With water: RCOCl + H₂O -> RCOOH + HCl (misty fumes, vigorous)
- With alcohol: RCOCl + R'OH -> RCOOR' + HCl
- With amine: RCOCl + 2R'NH₂ -> RCONHR' + R'NH₃Cl
- With NH₃: RCOCl + 2NH₃ -> RCONH₂ + NH₄Cl
- **SOCl₂** converts RCOOH to RCOCl (not HCl, not AlCl₃).

Ester Hydrolysis

- Acid hydrolysis: reversible, H₂SO₄ catalyst, heat under reflux. Gives RCOOH + R'OH.
- Alkaline hydrolysis: irreversible, NaOH, heat under reflux. Gives RCOONa + R'OH.

Amino Acids

- Draw the zwitterion: +H₃N-CHR-COO⁻. Do NOT remove H from any side chain.
- "Both dipeptides" means A-B AND B-A: draw both.
- Optical isomers: 3D diagrams with wedge and dash bonds. All amino acids except glycine are chiral.
- Isoelectric point: the pH at which the amino acid has zero overall charge.

Polymers

- Addition: from alkene monomers, no molecule lost, NOT biodegradable.
- Condensation: diol + diacid (polyester) or diamine + diacid/diacyl chloride (polyamide), loss of H₂O or HCl, CAN be hydrolysed.
- When drawing repeat units: show the bond extending from each end; for condensation polymers, check you have subtracted the H₂O lost.
- Polymer Mr: "Many candidates were successful with the first step but the best responses included the second step taking into account the Mr of water" (2019 ER). Subtract n-1 water molecules.

Phenol

- More reactive than benzene towards electrophiles because the lone pair on O is delocalised into the ring, increasing electron density.
- Reacts with Br₂(aq) without a catalyst (unlike benzene which needs a halogen carrier such as AlBr₃ or FeBr₃).
- Products are 2,4,6-trisubstituted (all positions ortho and para to OH).

Analytical Chemistry

- **¹H NMR**: consider adjacent protons on BOTH sides when predicting splitting. Equivalent CH₂ groups give singlets, not triplets. Must explain splitting with reference to the number of adjacent H, not just "n+1 rule."
- **¹³C NMR**: do NOT discuss splitting or relative peak areas. Only count distinct carbon environments. "Candidates struggled to recognise the two different carbon environments within the benzene ring" (2024 ER). Account for symmetry.
- **IR**: distinguish alcohol O-H (3200-3550, broad) from carboxylic acid O-H (2500-3300, very broad). "Some candidates assumed the unknown was a carboxylic acid, attributing the sharp peak just below 3000 cm⁻¹ to an OH group" (2023 ER).
- **Structure determination**: check that your proposed structure is consistent with ALL the data. "Many tried to make the data fit their proposed structure rather than the other way round" (2023 ER).
- **D₂O in NMR**: removes O-H and N-H peaks (exchangeable protons). If a peak disappears when D₂O is added, it was an O-H or N-H proton.

Practical

- **Reflux vs distillation**: reflux = heat under condenser pointing upwards (keeps volatile reagents in flask). Distillation = condenser angled downwards (collects product). "Candidates frequently confused the two techniques" (2023 ER).
- **Recrystallisation** purifies SOLIDS. **Separating funnel + distillation** purifies LIQUIDS. "Lots of confusion with purification with organic liquid was seen, such as references to use of a separating funnel, drying agents and distillation" for recrystallisation (2023 ER).
- **Purification of organic liquids**: separating funnel (wash with water/Na₂CO₃ to remove acid), identify correct layer using density, dry with anhydrous salt (MgSO₄ or CaCl₂), distil at correct boiling point.
- **Recrystallisation**: dissolve in minimum hot solvent, hot filter (remove insoluble impurities), cool slowly (pure crystals form), filter under vacuum, wash with cold solvent (remove soluble impurities).

PART 4: THE EASY MARKS CHECKLIST

Before you hand in your paper, check these **every time**:

- ☐ All curly arrows start from **lone pairs or bonds** (never atoms)
- ☐ All structures have the **correct number of H atoms** (no pentavalent carbons)
- ☐ Skeletal OR displayed formula used **consistently** (not mixed)
- ☐ Optical isomers described as **non-superimposable** mirror images
- ☐ Stereoisomers defined as same **structural** formula (not molecular)
- ☐ CIP priority uses **atomic number** (not Mr/mass)
- ☐ Markownikoff explained by **carbocation stability** (not product stability)
- ☐ NaBH₄/HCN racemic mixture: stated **planar carbonyl** AND **equal attack from both faces**
- ☐ Percentage yield: identified **limiting reagent** first
- ☐ Polymer Mr: subtracted **water lost** in condensation
- ☐ Structure determination: checked structure against **ALL** spectral data

These checks are worth an estimated 8-15 marks per paper.

PART 5: TRENDS AND TOPIC PATTERNS

What 2025 Paper 2 covered (less likely in identical form)

- Hydrocarbon naming, sigma/pi bonds, E/Z isomerism, electrophilic addition with Br₂ and HCl (Q16)
- Multi-step synthesis with percentage yield (Q17*)
- Addition and condensation polymers, polymer disposal, acid anhydride (Q18)
- Nucleophilic substitution mechanism (2-bromopropane + CN⁻), purification of organic liquid (Q19)
- Benzene delocalised model, phenol reactivity, Friedel-Crafts, nitration mechanism, directing effects, reduction of nitro to amine (Q20)
- Carbonyl tests (2,4-DNP, Tollens', Na₂CO₃, Cr₂O₇²⁻), NaBH₄ mechanism, multi-step synthesis flowchart (Q21)
- Amino acids: TLC, dipeptide structures, beta-keto acid decomposition (Q22)
- Combined analytical: IR, ¹³C NMR, elemental analysis, MS, ¹H NMR structure determination (Q23)

Topics absent for multiple years

Topic	Last appeared	Gap
Protein hydrolysis (peptide bonds)	2020	5 years
Friedel-Crafts as main mechanism Q	2024	1 year
Gas chromatography as main focus	2024	1 year
Acid anhydride as main focus	Never	Always brief (~1-2 marks)

Note: Diels-Alder (2021) and ozonolysis (2022) each appeared once as unfamiliar-context questions testing general curly arrow skills. They are not named specification topics and should not be revised as standalone content.

Topics that appear every year (never skip)

- Carboxylic acids and esters (9/9)
- Alcohols: oxidation, elimination, classification (9/9)
- Combined spectral analysis / structure determination (9/9)
- Multi-step organic synthesis (9/9)
- Aldehyde/ketone chemistry (9/9)
- IR spectroscopy (9/9)
- Mass spectrometry (9/9)
- Moles / stoichiometry calculations (9/9)

Topics in spec that have been tested only briefly

These are examinable but have never been the focus of a large question:

- Acid anhydrides (always 1-2 marks, never a dedicated question)
- Biodegradability of polymers (always ~1 mark)
- Benzene hydrogenation enthalpy as evidence for delocalisation (MCQ only, never structured)

What the trends point towards

Based on what 2025 covered heavily (and therefore is less likely in the same form) and what has been absent:

- **Protein/peptide hydrolysis** as a main question: last appeared 2020, 5-year gap. Amino acids appear 7/9 but the hydrolysis angle has been absent.
- **Friedel-Crafts acylation** as the main aromatic mechanism: 2025 focused on nitration. The paper tends to alternate between nitration and Friedel-Crafts.
- **Alkaline ester hydrolysis** (mechanism or comparison with acid): has appeared but not been a structured focus recently.
- **"Complete the mechanism" for an unfamiliar reaction**: OCR has increasingly given partial mechanisms since 2019. Expect at least one where the intermediate is provided and you add curly arrows.
- **¹H NMR with D₂O**: appeared 2020 and 2022. OCR likes testing whether students know D₂O removes O-H and N-H peaks.
- **Practical purification**: separating funnel + distillation for liquids, OR recrystallisation for solids. Examiners repeatedly flag confusion between the two.

Typical paper structure

The paper follows a consistent layout across all 9 years:

- **Q16:** large multi-part organic question (13-20 marks), usually alkenes or hydrocarbons
- **Q17-Q19:** one of these is always a Level of Response synthesis question (6 marks)
- **Q20-Q21:** the two biggest questions (typically aromatic chemistry and carbonyl chemistry, 15-22 marks each)
- **Q22:** usually amino acids (6-10 marks)
- **Final question:** always combined spectral analysis / structure determination (6-10 marks)

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